INCH-POUND

MIL-PRF-49147D(CR)
4 March 1999
SUPERSEDING
MIL-L-49147C(CR)
27 May 1992

PERFORMANCE SPECIFICATION

LENS ASSEMBLY, EYEPIECE
VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Lens Assembly Eyepiece, referred to herein as eyepiece, which is part of Viewer, Driver's Night Vision AN/VVS-2(V).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to insure the completeness of this list, document users are cautioned that they must meet all specified requirements in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document of by letter.

AMSC N/A FSC 5855

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2.1.2 Reference documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.c).

DRAWINGS

USA COMMUNICATIONS - ELECTRONICS COMMAND
SM-D-771327 - Lens Assembly Eyepiece

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The eyepiece is mounted to and is part of the Viewer, Driver's Night Vision AN/VVS-2(V).

3.2 Construction. The eyepiece shall be constructed to assure compliance with the contract configuration baseline and in accordance with SM-D-771327 as applicable.

3.3 First article testing (FAT). When specified in the contract or purchase order (see 6.2.e and 6.2.g) the contractor shall furnish eyepieces for FAT inspection and approval (see 4.2).

3.4 Materials. The contractor shall select parts and materials that are fully capable of meeting all operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but not mandatory. Selection criteria of the class, grade or type part shall ensure the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.5 Performance characteristics.

3.5.1 Diopter setting. The eyepiece shall be set at -2.0, +0.3/-0.1 diopter as viewed from a distance of 50 millimeters (mm) on axis when installed on the AN/VVS-2(V).

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3.5.2 Resolution. The eyepiece shall have a minimum resolution of 20 lp/mm when measured using a high contrast USAF 1951 Resolution Power Test Target (clear lines on black background or black lines on a clear background) and viewed binocularly from a 50 mm eye relief.

3.5.3 Magnification. The on-axis magnification shall be between 3.75 and 4.0.

3.5.4 Transmission. The transmission of the eyepiece shall be 80 percent minimum for the region of spectral output of a P-20 phosphor.

3.5.5 Field of view. The field of view of the eyepiece shall be 42.5 degrees minimum.

3.5.6 Field flatness. The field flatness shall deviate no more than 1/4 diopter over a 46 mm format with a 100 mm radius of curvature for the fiber optic magnifier output.

3.6 Environmental. Verification of conformance after environmental tests shall be Diopter setting (3.5.1), Resolution (3.5.2) and Transmission (3.5.4).

3.6.1 Temperature Extremes. The eyepiece shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to storage at +68°C or -51°C. The eyepiece shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to repetitive changes between +23°C and +68°C within 5 minutes and between +23°C and -51°C within 5 minutes. Soak time at each temperature extreme shall be sufficient for the eyepiece to attain temperature stabilization. Time for temperature stabilization is defined as the amount of time required for the most internal part of the unit under test to achieve the specified temperature.

3.6.2 Immersion. The eyepiece shall not be damaged (see 3.10.1) after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes. The eyepiece shall be pre-conditioned by exposure to a temperature of +45°C ±3°C for a period of 2 hours prior to being immersed in the fresh water at a temperature of +18°C ±5°C.

3.6.3 Vibration. The eyepiece shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to vibration equivalent to that encountered in tracked vehicles, during 5000 miles of operation. Vibration shall be limited to one sine sweep from 5.0 to 500 Hertz (Hz.) and back to 5.0 Hz. within 15 minutes in the vertical axis only. Vibration shall be performed from 5.0 to 5.5 Hz at 1 inch displacement, 5.5 to 30 Hz. at 1.5 g’s peak, 30 to 50 Hz. at 0.33 inch displacement and 50 to 500 Hz. at 4.2 g’s peak (see 3.10.5).

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3.6.4 Shock.

3.6.4.1 Basic shock. The eyepiece shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to basic shock of 40g's (see 3.10.5) at 18 milliseconds.

3.6.4.2 Weapon shock. The eyepiece shall not be damaged nor suffer degradation of performance (see 3.10.1 and 3.10.4) after being subjected to simulated weapon firing shock of 100 g's (see 3.10.5)at 1.5 milliseconds one half sine pulse.

3.7 Treatment and paintings. Treatment and painting shall be anti-reflective, shall be rust resistant and as specified on applicable drawings unless otherwise specified in the contract or purchase order.

3.8 Marking. Marking shall be in accordance with applicable drawings unless otherwise specified in the contract or purchase order.

3.9 Workmanship. Standards of workmanship shall be such that the eyepiece will meet all requirements of this specification and any referenced specifications or drawings.

3.10 Technical interpretations. The following technical interpretations are, when referenced in sections 3 or 4, mandatory for this specification.

3.10.1 Damage. Damage is defined as any breakage, loosening, shift, dents, failure of any finish, hardware, connection, or component, or condensation, fogging or moisture on any internal surface.

3.10.2 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

3.10.3 Environmental temperature. Environmental chamber temperature shall be controlled to within ±5°C of the specified temperature unless otherwise specified.

3.10.4 Degradation of performance. Any deterioration in quality, or failure to operate in accordance with the requirements of this specification.

3.10.5 "g". "g" is defined as an acceleration or deceleration of 32.17 feet per second per second.

4. VERIFICATION

4.1 Classification of inspection. Inspection shall be classified as follows:

a. First article tests (4.2)
b. Conformance inspection (4.3)

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4.2 First article testing. When specified in the contract or purchase order (see 6.2.g),(see 3.3) shall be performed by the contractor. The quantity of first article test samples shall be as specified in the contract or purchase order (see 6.2.e) and shall be selected from the first production lot.

4.2.1 FAT Test. Each eyepiece in the first production lot shall be subjected to the performance tests of Table I, which may be performed in any order . After completion of this testing, eyepieces selected as samples for FAT testing shall be subjected to the tests of Table II, which may be performed in any order with the exception that immersion shall be performed last. Failure of any test shall be cause for rejection of that eyepiece and may be cause for rejection of the FAT.

TABLE I. Unit first article testing.

Inspection	Requirement Paragraph
Diopter setting	3.5.1
Resolution	3.5.2
Magnification	3.5.3
Transmission	3.5.4
Field of view	3.5.5
Field flatness	3.5.6

TABLE II. Sample first article testing.

Inspection	Requirement Paragraph
Temperature Extremes	3.6.1
Immersion	3.6.2
Vibration	3.6.3
Shock	3.6.4

4.2.1.1 Disposition of FAT samples. Disposition of FAT samples shall be as specified in the contract or purchase order (see 6.2.e).

4.3 Conformance inspection. Requirements for conformance inspections shall be as specified in the contract or purchase order (see 6.2.f). Unless otherwise specified, all tests shall be performed at ambient temperature (see 3.10.2)

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2.h). When actual packaging of materials is to be perfumed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activity within the Military Department of Defense agency, or within the Military Department’s System Command. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CE-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The eyepiece covered by this specification is intended for use in the Viewer, Driver's Night Vision AN/VVS-2(V), an optical viewing device capable of directing the line of sight to the driver's eyes.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Necessary actions by contractor in the event of a lot failure.
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
d. Sampling plans for Group B and Group C testing.
e. Quantity and schedule of FAT and disposition of FAT samples.
f. Statement identifying that the contractor must submit his proposal for conformance inspection to include specific requirements to be verified and the sampling plan to be used to provide the desired confidence level.
g. Statement specifying that FAT testing is, or is not, required.
h. Packaging requirements.

6.3 Definitions. See 3.10.

6.4 Subject term keyword listing.

Diopter setting
Field flatness
Field of view
Magnification
Resolution
Transmission

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

Custodian:
Army - CR

Preparing Activity:
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Project 5855-0137

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49147D (CR)	**2. DOCUMENT DATE (YYMMDD)** 990304

3. DOCUMENT TITLE
Lens Assembly, Eyepiece Viewer, Driver's Night Vision, AN/VVS-2 (V)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9129 (2) AUTOVON 992-9129
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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